# FASHION RECOMMENDATION ON STREET IMAGES

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## ABSTRACT

Learning the compatibility relationship is of vital importance to a fashion recommendation system, while existing works achieve this merely on product images but not on street images in the complex daily life scenario. In this paper, we propose a novel fashion recommendation system: Given a query item of interest in the street scenario, the system can return the compatible items. More specifically, a two-stage curriculum learning scheme is developed to transfer the semantics from the product to street outfit images. We also propose a domain-specific missing item imputation method based on style and color similarity to handle the incomplete outfits. To support the training of deep recommendation model, we collect a large dataset with street outfit images. The experiments on the dataset demonstrate the advantages of the proposed method over the state-of-the-art approaches on both the street images and the product images.

***Index Terms—*** Fashion Recommendation, Outfit Completion, Street Photos, Curriculum Learning

## 1. INTRODUCTION

Due to the huge profits brought by online shopping, visual research on fashion becomes increasingly popular. However, most of the studies focus on the attribute detection [1] and recognition [2], clothing parsing [3] and landmark localization [4], fashion retrieval [5], *etc*. Recently, there is an increasing demand for developing an automatic fashion recommendation system. Such fashion recommendation system brings benefits to both the customers and online merchants, since it not only provides clear targets for customers, but also encourages them to purchase an outfit.

The key task of a fashion recommendation system is to learn the compatibility relationships. Such compatibility relationship can be learned among items from different categories. Veit *et al.* [6] proposed to explore the cross-category relationships with the Siamese network [7]. McAuley *et al.* [8] developed a distance metric on the deep features extracted from Amazon co-purchase data to measure the compatibility. He *et al.* [9] explored the localized notions of "relatedness" for compatibility evaluation. To improve the representation power, Veit *et al.* [10] developed a shared embedding combined with mask learning to capture the compatibility in different fascades. Vasileva *et al.* [11] further proposed to learn a non-linear embedding space for each category pair with triplet networks. However, these approaches only allow the evaluation of the compatibility between every two categories of items, without considering the overall look as a whole. Han *et al.* [12] tackled this problem by exploiting the sequence of co-occurring product images rather than pairs of items. Hsiao *et al.* [13] proposed to generate the personal wardrobe from the fashion images in different scenarios.

However, most of the above methods mainly focus on the product images and ignore the street images captured in more complex and realistic daily life scenario . Fashion recommendation based on street images is quite challenging in three aspects: 1) The cluttered background distracts the recommendation model learning from clothing items; 2) The densely annotated meta-data are not available; 3) The training data contain many incomplete outfits.

In this paper, we propose a novel fashion recommendation system, which is capable of returning the compatible items given a query item of interest in the street scenario. We consider the street image as a composition of individual product item and the context information. To exclude the distractive effect of cluttered background, a fashion apparel detector is developed to detect and crop each product item. To address the problem of sparsely annotated outfits in the street images, we propose to develop a curriculum learning scheme to transfer the semantics from rich annotated product images to street images. And for the incomplete outfits, we propose a novel domain-specific missing imputation method based on color and style similarity. Moreover, we construct a dataset by collecting street images from the social media and verify our proposed framework on this dataset.

Our contributions are twofold: (1) We develop a fashion recommendation system using street images, which efficiently learns from images with cluttered background,

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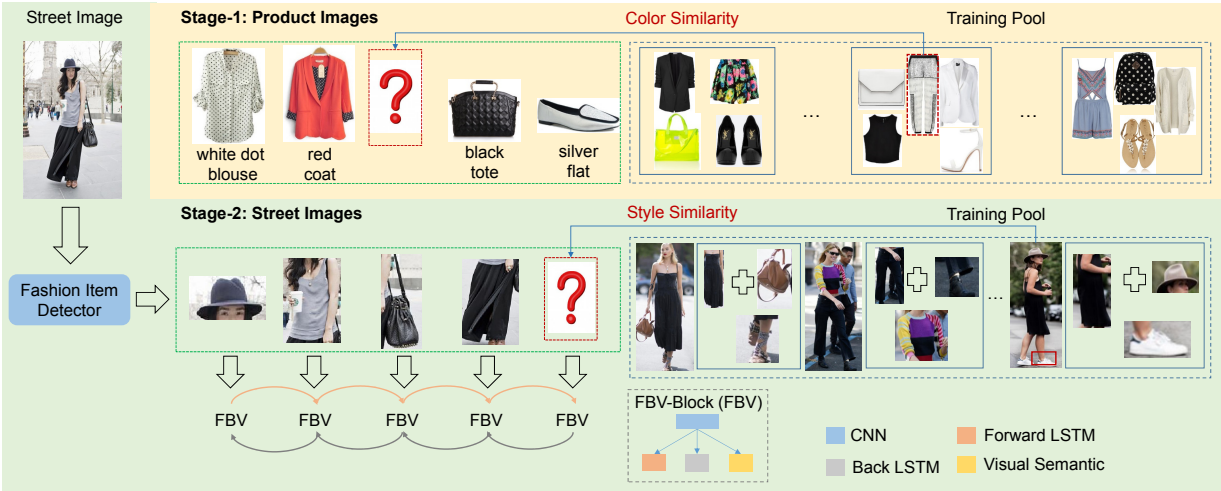


**Fig. 1**. The illustration for the pipeline of fashion recommendation system.

sparsely annotated and incomplete outfit items; (2) We build a large-scale dataset with street images in different occasions and regions.

## 2. PROPOSED METHOD

In this section, we will introduce a novel fashion recommendation system from the street images and the overall framework is illustrated in Fig. 1. The main function of this framework is to transfer the semantic knowledge learnt from the product images to the sparsely annotated street images via a two-stage curriculum learning strategy. Furthermore, to deal with the incomplete items (denoted as the red question mark in Fig. 1) in an outfit, we develop a domain-specific missing item imputation approach with style similarity on street images and color similarity on product images. The more details of each component will be introduced in the following paragraphs.

For a street outfit image $I_i$ in the street scenario, a fashion item detector, trained with the pixel-level annotations, is utilized to detect and crop the individual items. Then it is represented as a sequence of $m$ items as $I_i = [I_1^i, I_2^i, ..., I_m^i]$ in the pre-defined order (from top to bottom to accessories), which are subsequently fed into the Convolutional Neural Network (CNN) model for feature learning and grouped into a feature set, denoted as $\mathbf{X}_i = [\mathbf{x}_1^i, \mathbf{x}_2^i, \ldots, \mathbf{x}_m^i]$. To model the compatibility relationships among these items, a bi-directional Long Short Term Memory (Bi-LSTM) network is introduced into the proposed framework. For the $t$ th item $\mathbf{x}_t^i$ , a forward LSTM predicts the next image given previous images and its loss function $F_1(\mathbf{X}_i; \Theta_1)$ can be expressed as below:

$$F_1(\mathbf{X}_i; \Theta_1) = -\frac{1}{m}\sum_{t=1}^{m} \log p(\mathbf{x}_{t+1}^i | \mathbf{x}_1^i, \mathbf{x}_2^i, \ldots, \mathbf{x}_t^i; \Theta_1), \tag{1}$$

where $\Theta_1$ denotes the parameters of the forward prediction model. Similarly, the backward LSTM predicts the backward images in the reverse order and the loss function can be written as below:

$$F_2(\mathbf{X}_i; \Theta_2) = -\frac{1}{m}\sum_{t=m-1}^{1} \log p(\mathbf{x}_t^i | \mathbf{x}_{t+1}^i, \mathbf{x}_{t+2}^i, \ldots, \mathbf{x}_m^i; \Theta_2), \tag{2}$$

where $\Theta_2$ indicates the parameters of the backward prediction model. Similar to [12], to exploit the multi-modal similarity (*e.g.,* images and the associated text descriptions), we define a joint visual-semantic embedding space as follows:

$$\begin{aligned} F_3(\Theta_3) = &\sum_u \sum_s \max(0, m - d(u,v) + d(u,v_s)) + \\ &\sum_v \sum_s \max(0, m - d(u,v) + d(u_s,v)), \end{aligned} \tag{3}$$

where $\Theta_3$ means the parameters of visual-semantic model. $(u, v)$ is a paired image-text and $v_s$ is the non-matching text with $u$. Similarly, $u_s$ is the non-matching image to $v$. Thus, the fashion recommendation model can be trained by minimizing the sum of the above three loss functions denoted as below:

$$\min_{\Theta_1,\Theta_2,\Theta_3} F = \sum_i (\lambda_1 F_1(\mathbf{X}_i; \Theta_1) + \lambda_2 F_2(\mathbf{X}_i; \Theta_2)) + \lambda_3 F_3(\Theta_3), \tag{4}$$

where $\lambda_1$, $\lambda_2$, and $\lambda_3$ are the hyper-parameters that control the importance of forward, backward and visual-semantic losses,



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respectively, as shown in the FBV-Block (FBV) of Fig .1. In the proposed framework, we first learn the model parameters $\Theta_1$, $\Theta_2$ and $\Theta_3$ on product images and then transfer them on street images.

Due to the ineffectiveness of the Faster-RCNN [14] detector on small objects [15] and clothes in large deformation [16], there exist many incomplete outfits. According to the prior knowledge of fashion matching rules, we evaluate whether an outfit is complete or not based on the two criterions: 1) Upper body (top, coat) + Lower body (short, skirt, pant) + Accessories (handbag, hat, shoes); 2) Dress + Accessories (handbag, hat, shoes). Note that at least one of the accessories is required. Due to the discrepancy in the presence of text descriptions (*e.g.,* meta-data), we deal with the incomplete street and product images in different manners described as below.

For the street images, due to the lack of the densely annotated meta-data, the missing items are completed based on the style similarity, which is measured by the StyleNet [17]. For notation simplicity, we denote the style extraction network as $f_s(\cdot)$. The overall style is evaluated by the composition of the individual items and the context information. Given a style descriptor $f_s(\cdot)$, the style similarity between a pair of street outfits $I_i, I_j$ is calculated as below:

$$d(I_i, I_j) = \|f_s(I_i) - f_s(I_j)\|_2^2, \tag{5}$$

where $\|\cdot\|_2$ represents the $\ell_2$ norm and smaller $d(I_i, I_j)$ indicates the more similarity between $I_i$ and $I_j$.

For the product images, the incomplete outfits are completed based on the color similarity. The color similarity between a pair of items is categorized into three levels and each level will be assigned by a score $w_l, l = \{1, 2, 3\}$. Level 1 similarity indicates two colors are exactly the same. Level 2 similarity indicates that two colors are different but in compatible lists. That is, one of them is in black, white or grey color, which are suitable in most cases. Level 3 similarity indicates they are in different colors. And the color similarity score $F_c(i, j)$ between item $i$ and item $j$ is calculated as the score $w_l$ with different level, denoted as below:

$$F_c(i,j) = \begin{cases} w_1, & \text{Level 1 Similarity} \\ w_2, & \text{Level 2 Similarity} \\ w_3, & \text{Level 3 Similarity.} \end{cases} \tag{6}$$

Therefore, the pairwise similarity between a pair of outfits $I_i$ and $I_j$ in the product images, denoted by $C_{I_i,I_j}$, is computed as below:

$$C_{I_i,I_j} = \sum_{i}^{m_i} \sum_{j}^{m_j} F_c(i,j), \tag{7}$$

where $m_i$ and $m_j$ denote the number of items in polyvore outfits $I_i$ and $I_j$. And higher $C_{I_i,I_j}$ value indicates the more similarity between the pairwise outfits $I_i$ and $I_j$.

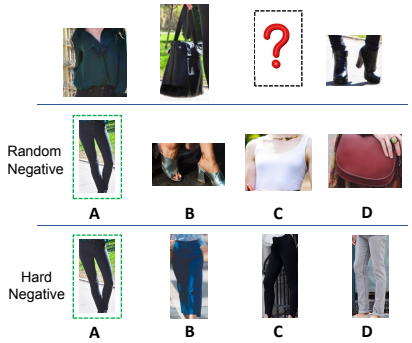


**Fig. 2**. Fill-in-the-blank evaluation metric. The matched ones are labeled in green bounding box.

In the test phase, given an item of interest in the street scenario, we first extract the embedding as the item representation, then it is fed forward into the Bi-LSTM to perform predictions in two directions. Finally, the output is a composition of items generated by forward and backward LSTM prediction.

## 3. EXPERIMENTS

### 3.1. Dataset and Experiment Settings

We collect the street outfit images from *Chictopia*[1] with occasion labels and *Wear*[2] with region-wise labels. Each item contains a product image and the outfit description. Finally, the dataset consists of 60,180 outfits in both asian (*e.g.,* China, Korea, Japan) and western (*e.g.,* UK and US) styles under the everyday scenario. For the product images, we utilised the Polyvore dataset [12] including 21,889 outfits with rich multi-modal information. After removing those images with irrelevant categories (*e.g.,* furniture, painting), 18,203 outfits are remained for training and evaluation.

**Experiment Setup and Evaluation Protocol.** Due to the lack of bounding box annotation on street images, we leverage the available fashion parsing dataset with pixel-level annotations as supervised information. The annotated training images are from three datasets: 1) Clothing Co-Parsing (CCP) Dataset [18] (1004 images); 2) Fashionista Dataset [19] (685 images); 3) Color-Fashion Dataset [20] (2682 images). Then they are further split into 4091 training and 280 testing images. The fashion apparel detector is built under the Faster-RCNN framework with the ResNet 50 as the backbone. The effectiveness of the apparel detector is based on the mean Average Precision (mAP).

For the Polyvore dataset, we conduct the experiments on 2,157 random negatives and 2,161 category-aware hard negatives. For the street dataset, 2,100 random negatives and

[1] http://www.chictopia.com
[2] http://www.wear.jp



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**Table 1**. Comparisons with state-of-the-art method for Random Negative (RN) and Hard Negative (HN) Examples.

| Approach | FITB Accuracy | | AUC Score | |
|---|---|---|---|---|
| | RN | HN | RN | HN |
| Bi-LSTM [12] | 0.73 | 0.68 | 0.80 | 0.84 |
| CSN [11] | 0.69 | 0.70 | **0.94** | **0.92** |
| Proposed | **0.77** | **0.72** | 0.83 | 0.86 |

**Table 2**. Evaluating the effect of the Curriculum Learning (CL) and the Missing Item Imputation (MII) on the street (the first three rows) and Polyvore dataset (the last two rows).

| Dataset | Approach | FITB Accuracy | | AUC Score | |
|---|---|---|---|---|---|
| | | RN | HN | RN | HN |
| Street | Proposed w/o MII | 0.74 | 0.69 | 0.81 | 0.84 |
| | Proposed w/o CL | 0.75 | 0.70 | 0.82 | 0.86 |
| | Proposed | **0.77** | **0.72** | **0.83** | **0.86** |
| Polyvore | Proposed w/o MII | 0.73 | 0.44 | 0.93 | 0.94 |
| | Proposed | **0.78** | **0.46** | **0.95** | **0.96** |

2,100 category-aware hard negatives are chosen for experiments. In the random negative examples, no prior knowledge of the missing blank (category) is provided; The hard negative indicates that the category of the missing item is known, the potential candidates can only be chosen in the same category. For the Polyvore dataset, the recommendation is pre-trained on the InceptionV3 model. For the hyper-parameters of Bi-LSTM model, we follow [12] as the guideline.

For quantitative evaluation, two metrics are utilized to estimate the performance of the fashion recommendation system, 1) FITB (fill-in-the-blank) accuracy, as shown in Fig. 2. Given multiple query items in an outfit (one item is erased, illustrated as the question mark), the system is expected to choose the right answer out of 4 choices; 2) fashion compatibility prediction accuracy, which is evaluated by the area under the ROC curve (AUC).

### 3.2. Experimental Results

For the fashion apparel detector, we set the correct detection Intersection over Union (IoU) threshold to 0.5. Our detector achieves 74.8% mAP score on the test images. For the fashion recommendation model, we compare the performance of the proposed method with two state-of-the-art algorithms: 1) sequence-based Bi-LSTM network [12] and 2) metric learning based CSN network [11].

We can find that the experiments conducted on the street dataset demonstrate that our proposed method achieved better performances in terms of FTIB accuracy with the other two state-of-the-art methods and comparable AUC scores.

**Fig. 3**. Fashion recommendation examples given the query item from the street scenario. Best view in color.

Fig. 3 demonstrate the recommended outfits with the query items in the street scenario. It can be seen that the proposed system is capable of recommending the outfits in variable styles and the overall styles of different fashion items in an outfit match with each other. Compared with the other two approaches, our method puts an emphasise on the overall style of the outfits rather than merely the color similarity.

### 3.3. Effectiveness of the Curriculum Learning and Missing Item Imputation

We evaluated the effectiveness of the curriculum learning and missing item imputation component in the model learning, as shown in the first three rows of Table. 2. It verifies the effectiveness of transferring semantics from the product images with densely annotated meta-data. Also, the imputation strategy based on the style similarity also improves the performance of recommendation. The results on the Polyvore dataset in the last two rows of Table 2 further verify the effectiveness of the imputation strategy on the product images.

## 4. CONCLUSION

In this paper, we propose a novel fashion recommendation system to return the compatible items given a query item in the street scenario. We transfer the high-level semantics from product images to sparsely annotated street images via the two-stage curriculum learning strategy. To deal with the incomplete outfits, we employ the style and color similarity as the guideline to perform the missing item imputation. Moreover, we build a large-scale dataset with street images in different occasions and region. The experimental results demonstrate the advantages over the state-of-the-art approaches.



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